Lecture 10: Evolution, Emotion, and Reason: Evolution and Rationality (full transcript)

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Overview: This lecture introduces students to the study of psychology from an evolutionary perspective, the idea that like the body, natural selection has shaped the development of the human mind. Prominent arguments for and against the theory of natural selection and its relationship to human psychology are reviewed. Students will hear several examples of how studying mental phenomenon from an evolutionary perspective can help constrain theories in psychology as well as explain many prevalent human instincts that underlie many of our most basic behaviors and decisions.

Chapter 1. The Modern Biological Account of the Origin of Psychological Phenomena [00:00:00]

Professor Paul Bloom: We began the course by talking about one of the foundational ideas of modern psychology. This is what Francis Crick described as "The Astonishing Hypothesis," the idea that our mental life, our consciousness, our morality, our capacity to make decisions and judgments is the product of a material physical brain. What I want to talk about today and introduce it, and it's going to be a theme that we're going to continue throughout the rest of the course, is a second idea which I think is equally shocking, perhaps more shocking. And this has to do with where mental life comes from, not necessary its material nature, but rather its origin. And the notion, this other "astonishing hypothesis," is what the philosopher Daniel Dennett has described as Darwin's dangerous idea. And this is the modern biological account of the origin of biological phenomena including psychological phenomena.

Now, people have long been interested in the evolution of complicated things. And there is an argument that's been repeated throughout history and many people have found it deeply compelling, including Darwin himself. Darwin, as he wrote The Origin of Species, was deeply persuaded and moved by this argument from — in the form presented by the theologian William Paley. So, Paley has an example here. Paley tells — gives the example of you're walking down the beach and your foot hits a rock. And then you wonder, "Where did that rock come from?" And you don't really expect an interesting answer to that question. Maybe it was always there. Maybe it fell from the sky. Who cares? But suppose you found a watch on the ground and then you asked where the watch had come from. Paley points out that it would not be satisfying to simply say it's always been there or it came there as an accident. And he uses this comparison to make a point, which is a watch is a very complicated and interesting thing.

Paley is — was a medical doctor and Paley goes on to describe a watch and compare a watch to the eye and noticing that a watch and the eye contain multitudes of parts that interact in complicated ways to do interesting things. In fact, to change and to update the analogy a little bit, an eye is very much like a machine known as a camera. And they're similar at a deep way. They both have lenses that bend light and project an image onto a light-sensitive surface. For the eye the light-sensitive surface is the retina. For the camera it's the film. They both have a focusing mechanism. For the eye it's muscles that change

the shape of the lens. For a camera it's a diaphragm that governs the amount of incoming light. Even they're both encased in black. The light-sensitive part of the eye and part of the camera are both encased in black. The difference is — So in fact, the eye and a camera look a lot alike and we know the camera is an artifact. The camera has been constructed by an intelligent — by intelligent beings to fulfill a purpose.

In fact, if there's any difference between things like the eye and things like a camera, the difference is that things like the eye are far more complicated than things like the camera. When I was a kid I had this incredible TV show called "The Six Million Dollar Man." Anybody here ever seen it or heard of it? Oh. Anyway, the idea is there's a test pilot, Steve Austin, and his rocket jet crashes and he loses his — both legs, his arm and his eye, which sounds really bad but they replace them with bionic stuff, with artificial leg, artificial arm and an artificial eye that are really super-powered. And then he fights crime. [laughter] It was [laughs] really the best show on. It was really good, [laughter] but the thing is this was in 1974. It's now over thirty years later and it's true then and it's true now, this is fantasy. It doesn't make it to the level of science fiction. It's fantasy. We are impossibly far away from developing machines that could do this. We are impossibly far away from building a machine that can do what the human eye does. And so somebody like Paley points out, "Look. The complexity of the biological world suggests that these things are complicated artifacts created by a designer far smarter than any human engineer. And the designer, of course, would be God."

I went to Goggle Images. That — I don't mean that to be sacrilegious [laughter] in any sense. You could try this. I went to "Google Images" and typed in "God" and this [a picture of an old-bearded man wearing a crown] is what showed up right in the middle so — And this, Paley argued, and it was — has been convincing throughout most of history, is a perfectly logical explanation for where these complicated things come from. It also has the advantage of being compatible with scripture and compatible with religious beliefs, but Paley made the point this stands on its own. If you find complicated things that — complicated artifacts, you don't assume they emerged by accident. You assume that they were created by an intelligent being.

Now, this view has always had problems. This view, you could call it "creationism," which is that biological structures were created by an intelligent being, has always had problems. One problem is it pushes back the question. So you ask, "Where did that intelligent being come from?" And this is a particularly serious problem from the standpoint of the evolution of psychological structures. So, we want to know, "how is it that creatures came across — upon this earth with the ability to reason and plan and do things?" And then the answer is "well, another creature with that ability created us." That doesn't necessarily mean it's wrong, but it means it's unsatisfying. You immediately want to get an explanation for where that other creature comes from.

More to the point, there's always been evidence for evolution. And what I mean by evolution here isn't necessarily a specific mechanism, but merely the fact that body parts like the eye didn't emerge all of a sudden, but rather have parallels both within other existing animals and across human and biological history. This evidence comes in different forms. There is fossil evidence for different body parts suggesting that they have evolved from more rudimentary form. There is vestigial characteristics. And what this means is there are characteristics that human bodies have that are somewhat inexplicable,

like the human tailbone or goose-bumps, unless you view them — the human body in its current form as modifications from a previous form.

There are parallels with other animals. And this is clear in psychology. So, a human brain is different from the rat, cat, and monkey brain but at the same time you see them following a sort of common plan and common structures. And one rational inference from this is that they're linked through evolutionary descent.

Finally, there is occasional poor design. So, Paley rhapsodized about the remarkable powers of the human body and the different body parts, but even Paley admitted that there are some things which just don't work very well. Your eye contains a blind spot because of how the nerves are wired up. In the male urinary system the urethra goes through the prostate gland instead of around it, which leads to many physical problems in men later on in life. And so you're forced to either argue that these are really good things or that God is either malicious or incompetent. And those are difficult arguments to make.

So, these are problems with the creationist view. But still, for the longest time in human intellectual history there was no alternative. And in fact, Richard Dawkins, the most prominent evolutionary — one of the most prominent evolutionary biologists alive and one of the most staunchest critics of creationism, has written in The Blind Watchmaker saying, look, anybody 100 years ago or 150 years ago who didn't believe that God created humans and other animals was a moron because the argument from design is a damn good argument. And in the absence of some other argument you should go — defer to that. You should say, "Well, there are all of these problems but humans and other biological forms must have divine creation because of their incredible rich and intricate structure." What changed all that of course was Darwin. And Darwin — Darwin's profound accomplishment was showing how you get these complicated biological structures, like the eye, emerging through a purely non-intentional, non-created process, a purely physical process. And this could be seen as equal in importance to the claim that the Earth revolves around the Sun and that we're not the center of the universe. And in fact, some scholars have made a suggestion which seems plausible, that the idea of natural selection is the most important idea in the sciences, period.

So, this is not a course in evolution and I expect people to have some background. If you don't have a background in it, you could get your background from external readings but also from — the Gray textbook and the Norton readings will both — will each provide you with enough background to get up to speed. But the general idea is that there are three components to natural selection. There is variation. And this variation gives rise to different degrees of survival and reproduction and gets passed on from generation to generation and gives rise to adaptations, what Darwin described as "that perfection of structure that justly excites our imagination."

And the biological world has all sorts of examples. You look at camouflage. Prior to Darwin one might imagine that some intelligent creator crafted animals to hide from their prey. But now we have a different alternative, which is that animals that were better hidden survive better, reproduce more, and over the course of thousands, perhaps millions of years, they've developed elaborate camouflage. There's been a lot of work on Paley's favorite example – the eye. So Darwin himself noted that the human eye did not seem to emerge all at once but rather you could look at other animals and find parallels in other animals that seem to suggest that more rudimentary forms are possible. And more

recently computer simulations have developed — have been developed that have crafted eyes under plausible assumptions of selective pressure and what the starting point is.

So, this is the theory of natural selection. The good question to ask is, "why am I talking about evolution in Introduction to Psychology class?" And the answer is that there are two ideas which come together. And in fact, they're both of the dangerous ideas. One idea is that Darwin's idea — that biological forms evolve through this purely physical process. The second idea, the rejection of Descartes, is that our minds are the product of physical things and physical events. You bring these together and it forces you to the perspective that what we are — our mental life is no less than the eye, no less than camouflage, the product of this purely physical process of natural selection. More to the point, our cognitive mechanisms were evolved not to please God, not as random accidents, but rather for the purpose of survival and reproduction. More contentiously, you could argue they've been shaped by natural selection to solve certain problems. And so, from an evolutionary point of view, when you look at what the brain is and what the brain does, you look at it in terms of these problems. And this is what psychology is for. This is what our thinking is for. We have evolved mental capacities to solve different problems: perception of the world, communication, getting nutrition and rest, and so on.

Chapter 2. Avoiding Misconceptions When Applying Evolutionary Theory to Psychology [00:13:35]

Now, we're going to talk about how to apply evolutionary theory to psychology. But as we're doing so we have to keep in mind two misconceptions. There are two ways you can go seriously wrong here. The first is to think that, well, if we're taking an evolutionary approach then natural selection will cause animals to want to spread their genes. So, if we're being biological about it, that means everybody must run around thinking "I want to spread my genes." I want to — and this is just really — Oops. I shouldn't do that. This is really wrong. It's [the text on the slides] even in red. And what this fails to do is make a distinction between ultimate causation and proximate causation. And those are technical terms referring to — Ultimate causation is the reason why something is there in the first place, over millions of years of history. Proximate causation is why you're doing it now. And these are different. Obviously, for instance, animals do all sorts of things to help survive and reproduce but a cockroach doesn't think "oh, I'm doing this to help survive and reproduce and spread my genes." A cockroach doesn't know anything about genes. Rather, the mechanisms that make it do what it does are different from its own mental states, if it has any — why it does them.

This is a point nicely made by William James. So, William James is asked, "Why do we eat?" And he writes,

Not one man in a billion when taking his dinner ever thinks of utility. He eats because the food tastes good and makes him want more. If you asked him why you should want to eat more of what tastes like that, instead of revering you as a philosopher, he will probably laugh at you for a fool.

And it's really the common sense answer. "Why are you eating?" Nobody's going to answer, "Because I must sustain my body so as to spread my genes in the future." Rather, you eat because you're hungry.

Those two theories, you eat because you're hungry and you eat to sustain your body so you could spread your genes in the future, are not alternative. Rather, they're different levels of explanation. And

you can't confuse them. The ultimate level which does appeal to survival and reproduction does not — is independent from the psychological level. To give another example, people protect their children so you ask, "Why do people protect their children? Why would somebody devote so much effort to protecting and helping and feeding their children?" Well, the evolutionary explanation is animals that don't protect their offspring don't last over evolutionary time. We protect our offspring because they contain fifty percent of our genes, but that's not the psychological explanation. Nobody but a deranged psychologist would ever answer, "Oh, I love my children because they contain fifty percent of my genes." Rather, the psychological explanation is a deeper — is different and has a different texture. And this will be a lot clearer when we talk about the emotions, where you could really see a distinction between the question of why we feel something from an evolutionary point of view and why we feel it from a day-to-day point of view.

The second misconception is that natural selection entails that everything is adaptive, that everything we do, everything we think is adaptive. This is wrong. Natural selection and evolution, more generally, distinguish between adaptations and byproducts and accidents. Many of you are currently, or will as you get older, suffer back pain. If I was to ask you, "So, why do you suffer back pain? How does back pain help you survive and reproduce?" Well, the answer is it's not an adaptation. Back pain is an accidental byproduct of how our backs are shaped. Don't go looking for an adaptive reason for hiccups or self-pity or bloating after you eat. There's all sorts of things a body will do that have no adaptive value, rather just accidents. We have a body that does all sorts of things. Some things it will do by accident and this is certainly true for psychology.

So, a lot of the things, for instance, that occupy our interest or our fascination in day-to-day life are almost certainly evolutionary accidents. The number — The three — Three of the main preoccupations of humans are pornography, television, and chocolate but if I asked you, "Why do you like porn?" and you'd say, "Because my ancestors who liked porn reproduced more than those who didn't," [laughter] it's not true. Rather, you like porn, assuming you do, [laughter] as an accident. You have evolved — For instance, should you be a heterosexual male, you have evolved to be attracted to women. That is most likely to be an evolutionary adaptation because being attracted to women and wanting to have sex with women is one step to the road to having kids, which is very good from an evolutionary perspective.

It so happens, though, in our modern environment that people have created images that substitute. So, instead of actually going out and seeking out women you could just surf the web for hours and hours and watch dirty movies and read dirty books – evolutionary adaptive dead ends. They're accidents. Why do you like chocolate bars, assuming that you do? It is not because your ancestors in the African savanna who enjoyed chocolate bars reproduced more than those who didn't. Rather, it is because we've evolved a taste for sweet things. And we've evolved a taste for sweet things, in part, because the sweet things in our natural environment like fruit were good for us. In the modern world we have created things like chocolate, which are not so good for us but we eat anyway.

A lot of the debates — There's a lot of controversy in psychology over the scope of evolutionary explanations. And a lot of the debate tends to be over what's an adaptation and what isn't. There are some clear cases. We have color vision. Why do we have color vision? Well, I think everybody would agree we have color vision as an adaptation because of the advantages it gives us for seeing and making visual distinctions. We are afraid of snakes. We're going to talk about that in more detail but there's a

straightforward adaptive story about that. We are afraid of snakes because, really, our ancestors who weren't afraid of snakes didn't reproduce as much as those that were. We like chocolate bars and we enjoy NASCAR. Those cannot be adaptations because chocolate bars and NASCAR are recent developments that could not have been anticipated by evolution.

Those are easy questions. Here are some hard questions. Music. Everywhere in the world people like music. Is this an adaptation for some selective advantage or is it an accident? Steven Pinker, who wrote The Language Instinct that you read before, caused a huge amount of controversy when he argued that music is just an evolutionary accident. He described it as auditory cheesecake, something we like to gorge ourselves on that have no — has no adaptive advantage. Other people argue music does have an adaptive advantage. Sometimes males use violence to coerce sex. Is male sexual violence a biological adaptation or is it an accident? There's more than one language. Is that just an accidental byproduct of the way language works or is there some sort of group or selectionist advantage sketched out in some way of having multiple languages? What about visual art? What about fiction? What about our love for stories? Those are all matters of heated debate.

And so, we have to keep in mind some things plainly are accidents. Some things almost certainly aren't accidents. Where the action is in the study of psychology and the study of evolution of cognition is trying to figure out which is which. So, those are the misconceptions we have to avoid. But still, who cares? Again this is an Introduction to Psych course. Why are we talking about evolution? Why should it matter to a psychologist how the mind has evolved? I'm going to talk about evolution now but for the rest of the course I'm just interested in how our minds are, period. So, why would evolution matter?

Chapter 3. Claims against the Evolutionary Psychology [00:22:39]

Well, many people think it doesn't. For instance – and they think it doesn't for different reasons – one claim is a metaphysical one. You might be a dualist. You might reject the idea your mental life is the product of your brain and hence evolution is irrelevant to psychology because the brain and the mind — because the brain, which may have evolved, has nothing interesting to do with the mind. Lisa Simpson got it wrong when she said the Pope — She got it half right when she said the Pope favored evolution. It is true. John Paul II, many years ago, made a statement saying that Darwinian theory is not incompatible. Darwinian theory is a view about the evolution of species that is not motivated by any animus, is a genuine scientific theory, and should it turn out to be true, it is not incompatible to truth about man as taught by the Church. And scientists were thrilled by this and they were — they said he's endorsing evolution. But what a fewer people talk about is the fact that after he said this he drew the line. He allowed for evolution of the body but he would not allow for evolution of the mind. So it was — he wrote:

If the human body takes its origin from preexisting living matter, the spiritual soul is immediately created by God. Consequently, theories of evolution which consider the mind as emerging from the forces of living matter or as a mere epiphenomenon of this matter are incompatible to the truth about man.

So, you might not want evolution to be true about the mind because you might believe that the mind is not subject to the same physical laws as the rest of the physical world. That's one way you could reject evolutionary psychology. Another way to reject evolutionary psychology is to accept that the mind is a

physical thing but then argue that all of these instincts and these hard-wired facets of human nature might exist for other animals but they don't exist for people. So, the anthropologist Ashley Montagu in '73, close to when The Six Million Dollar Man was shown, by the way, said:

With the exception of the reactions of infants to sudden withdrawals of support and to sudden loud noises, the human being is entirely instinctless. Man is man because he has no instincts, because everything he is and has become he has learned from his culture, from the man-made part of the environment, from other human beings.

You might say, "Look. He could believe that in '74 but, of course, all of the infant studies that have come out since then suggested that's not true and nobody would believe that nowadays." But in fact, the view is often hold — held — Louis Menand in a New Yorker article a few years ago wrote, "Every aspect of life has a biological foundation in exactly the same sense, which is that unless it was biologically possible it wouldn't exist. After that it's up for grabs." And this is in the context of an argument that evolution can't tell us anything about what's most interesting about people. Menand is not — is an educated, intelligent scholar. He is presumably well aware of the findings of Spelke and Baillargeon about how people are hard-wired to understand the objects in social life and so on. But his point is just that when it comes to the more interesting aspects of human nature, the stuff we're naturally, intuitively interested in, that's more cultural. And the evolutionary theory and Darwinian theory just doesn't have anything much to say about it, not because the mind is separate from the brain but just because humans are much more cultural organisms, and so biology has little to say about it.

There's a third objection, which is you might think, "Okay, the human mind actually does contain instincts. There is a human nature but we should just study it by studying people. How could evolution, the study of evolution, the consideration of evolution tell us anything interesting?"

Chapter 4. Ways in Which Evolution Helps Describe the Mind [00:26:42]

I actually, in my own work, think evolution can tell us some interesting things. And I want to try to make a case for ways in which evolution can inform and enlighten us about the mind as it is.

First, I want to make a point, which is although this course is Intro Psych and it is about the mind as it is, still I think by any account the evolution of consciousness, morality and so on, just is intuitively interesting. It's the sort of thing that people are just fascinated by and I think it's a question of interest in and of its own right. But here's how it could tell us about psychology. For one thing, it can tell us what can be innate and what cannot. So, some problems, some evolutionary problems, have been around for a long time and could lead to special biological adaptations. If I told you there is a biological adaptation for talking, mate selection, childcare; maybe it's true, maybe it isn't, but it's not crazy. From an evolutionary point of view, it's a reasonable possibility that it is.

Other problems are recent and our brains could not be specialized to deal with them: written communication, interacting with strangers, driving a car, playing chess. If you were to argue that there's a part of the brain devoted to playing chess, I would say you're utterly wrong. You cannot be right because, from an evolutionary point of view, there could be no such part of the brain evolved because playing chess is a recent innovation. As a result, a focus on evolution could help discipline us to make coherent claims about what is built-in and what isn't built-in.

Third, we're going to talk about human differences in this course. We're going to devote a class to human differences of the sort of what makes you different from her, different from her. Why do we have different intelligences in this class? Why are some of us arrogant and some of us humble? Some of us like — attracted to men, others attracted to women, and so on. But there's also questions of group differences. And evolutionary theory can help us say intelligent things about what sort of group differences you should expect because evolutionary theory predicts that some populations should evolve in different ways than others.

The most obvious example is that children should be different from adults. The evolutionary problems faced by a child are very different from the evolutionary problems faced by an adult. And you can make specific and rather interesting predictions about how children's brains should different — differ from adults' brains. Evolutionary theory predicts — does not make any predictions about racial differences or ethnic differences. Some might exist, but there's no adaptive reason why humans who have evolved in different parts of the world should have profound differences in their mental capacities.

What does evolutionary theory say about sex differences? Well, it says some interesting things, and we're going to devote a class to discussing them, but what I think is going to be true — proved to be important is that we'll be able to use evolutionary biology to talk sensibly about what sort of distinctions between the sexes, between males and females, one would expect to find and what sort one wouldn't expect to find. We can make educated predictions. I'm going to have — I want to put here a clip of a man. This is a scene from a movie, the movie "Roger Dodger," that begins with a man making quasi-evolutionary claims about the differences between men and women. And I want to put this as an example of what you could call "barroom evolutionary psychology." And I want us to hold this in our minds because we're going to return to these claims and discuss their validity. [clip playing]

I like this for a few reasons. First, I like the backward reference to William James and utility. Second, it is a gorgeous combination of some things that are actually reasonably rational and total bull crap. And — but what evolutionary biology will give us is the tools to distinguish between the two. On the face of it immediately, the ability to read maps, the claim that that has a biological — that differences in that ability have a biological root is crazy. On the other hand, the claim that one — that males may develop a trait not because it's advantageous but to attract females is less crazy. The telepathic stuff is really crazy but — ;So, I'm not at this point — We're going to devote a lecture to sex. I do not, at this point, want to make any claims one way or another. But what I want to suggest is that from a biological point of view we could say sensible and intelligent things about what differences should exist and what shouldn't.

Finally, and most of all, looking at something from the perspective of design, the perspective of what's it for, can often give you interesting insights as to its current nature. And I'll give you two quick examples, one that's not from psychology, one that is. Women suffer — Often women who are pregnant early in their pregnancy suffer from morning sickness, nausea, throwing up and so on. This has traditionally been viewed as just a breakdown in the system — too much hormones, everything's askew; women get nauseous. Margie Profet suggested an alternative and this won her theMacArthur Genius Award. And this was the claim that maybe pregnancy sickness is not an accident; rather, it's designed, it has a biological purpose. In particular, as the baby develops in the uterus, it is vulnerable to various sorts of poisons or teratogens. Profet suggested that pregnancy sickness is a hypersensitive

period where women are extremely sensitive, get extremely nauseous towards the sorts of foods that could damage their baby.

Now, if she just ended there it's a story. How do we know it's true? But then she went on to examine it the same way that any scientist examines any claim – by making predictions and testing them. And this makes some interesting predictions. It suggests the timing of onset and offset of pregnancy sickness, of morning sickness, should correspond to the period of maximal vulnerability on the part of the developing embryo or fetus. Suggested the types of foods avoided should correspond to those sorts of foods that were most deadly for the fetus and that were deadly for the fetus during the periods where humans evolved. This last qualification is an important one. Women do not develop an aversion to alcohol during pregnancy even though alcohol is extremely dangerous to the developing child. The answer is an easy one. Alcohol wasn't around during our evolutionary history and we could not have evolved a system to protect ourselves from it.

And finally, there should be a relationship between miscarriage and birth defects in a surprising direction. For Profet, and she has evidence to back this up, pregnancy sickness is not a glitch in the system. Rather, it's the sign of a healthy act of protective mechanism going on. And in fact, the more morning sickness the more the baby should be protected. Something which, by and large, appears to be true. That's an example of how the question — when dealing with this they say, "Hey. Women throw up when they get pregnant" and then say, "Look. Maybe that's not just a glitch. What's it for?" You could then learn some interesting things.

Here's a different example based on the last lecture, this wonderful lecture by Peter Salovey on sex and love where he talked about the "big three." These are the "big three" to remind you of what attracts us to somebody else. You are very attracted to the person next to you or a person that — because of proximity, similarity, familiarity. And there is abundant evidence supporting the truth of this. It's almost always true but the evolutionary psychologist looks at this and says there's something seriously wrong here. There are some cases where that has to be totally, absolutely mistaken. To realize what this is, think for a moment. What humans are you most close to, most similar to and most familiar with? What humans did you spend over ten years of your life with who are genetically and environmentally as close to you as if they were related, who you are intimately familiar with? Are those the humans that you find the hottest? [laughter] No. They're your siblings and they are not hot. [laughter]

I was on Google Images this morning. I put up some hot siblings and — but — although we may find them hot, they do not typically, with some rare and bizarre exceptions, find [laughter] one another hot. Why not? Well, this is not a huge puzzle from the standpoint of evolutionary biology. Evolutionary biology posits that humans, as well as other animals, should have incest avoidance. We should love — we should be attracted to those familiar to us, similar to us, close to us, but not kin. Kin are off limits. There is a good reason why. Because if you inter-mate with your kin you have bad offspring [laughter] and so animals should be wired up not to mate with their kin. And in fact, this is what happens.

There are — Parents of teenagers have all sorts of concerns. And a lot of the concerns are, in fact, sexual. How do you keep your son and/or your daughter from going out and having sex with too many people, or the wrong people, or unprotected sex? But there are no parenting guides in the world that

say "How do you keep your children from having sex with one another?" [laughter] You typically do not need to because they do not want to have sex with one another.

Now, this is — actually also illustrates the difference between proximate and ultimate causation. So, you think for yourself, "Eew. Do I want to have sex with my sister?" You don't think to yourself, "I would prefer not to, for the offspring that we will create will be nonviable and it'll be a waste of my reproductive efforts." Rather, you think, "Eew," because at a gut level you respond. And this sort of instinctive response is what you get from an evolutionary analysis of sex.

But this story is deeply incomplete because the question that gets raised is "how do you tell?" You don't want to have sex with your kin but how do you tell your kin? People don't carry their DNA markers on strips that you could see. How do you tell who your kin are? And this actually turns out to be a really interesting question. It used — And some research suggests that the answer is simple. You avoid sex with people you grew up with. And these studies actually come from kibbutz studies, studies where people are raised communally on an Israeli kibbutz. They know they're not related, but still, the fact that they were raised together as kids suggests that there's a cue at a gut level not to be attracted to one another.

It turns out there's some reason now to believe this story is incomplete. A paper that came out in Nature five days ago reported a series of extremely interesting studies. And they found that the cue of being raised together as a child with somebody — yes, that does diminish sexual desire, but an even bigger cue was "did you observe your parents, and in particular, your mother, taking care of that person?" If you did, that seriously diminishes sexual desire and brings it down to the level of disgust. And again, these are the sort of questions and issues you begin to ask when you approach things from an evolutionary perspective. Okay.

Chapter 5. Heuristics: Framing Effects, Base Rates, Availability Bias and Confirmation Bias [00:39:48]

For this lecture — the rest of this lecture and then the next couple of lectures, I'll be discussing some basic aspects of human nature that are, to some extent or another, informed by evolutionary theory. And what I want to start for the remainder of this lecture is a discussion of rationality. Now, some of you maybe not want to go into — not want to go into psychology because there's no Nobel Prize for psychology. You might all think, "Hey, if I'm going to go into the sciences I want a Nobel Prize. Think how proud Bubby and Zadie would be if I won a Nobel Prize. Wouldn't that be the best?" You can get one. Psychologists have won the Nobel Prize. Most recently, Danny Kahneman won a Nobel Prize. You win it in economics, sometimes medicine; not a big deal. He won it for his work done over the course of many decades on human rationality. And this work was done in collaboration with Amos Tversky, who passed away several years ago. And this work entirely transformed the way we think about human decision-making and rationality.

Kahneman and Tversky caused a revolution in economics, psychology, and the social sciences more generally, by causing us to shift from the idea that we're logical thinkers, who think in accord with the axioms of logic and mathematics and rationality, more towards the idea that we actually have sort of rough and ready heuristics. These heuristics served us well during the time — during our evolutionary

history, but sometimes they can lead us astray. And I want to give some examples of these heuristics. And I'll give four examples of heuristics that are argued to permeate our reasoning.

The first is "framing effects." This was a classic study by Kahneman and Tversky involving this sort of question. The U.S. is preparing for the outbreak of a disease that's going to kill six hundred people. There are two programs. Program A: If you follow it two hundred people will be saved. Program B: There's a one-third chance everybody will be saved and a two-third chance nobody will be saved. Who would choose program B? Who would choose program A? Okay. And that fits the responses. Most people choose program A. That's — It could go either way. What's interesting is if you frame the question differently, like this, you get very different responses. And instead of focusing on the people who will be saved, you focus on the people who will die and, instead of focusing on the chance that nobody will die and the chance that everybody will die, you'd flip it around, you get a corresponding flip. And this is known as a "framing effect."

The idea of a framing effect is that you could respond differently to a situation depending on how the options are framed. And, in particular, this combines with "loss aversion." People hate a certain loss. "Four thousand of these people will die" is extremely aversive and so the framing can influence your decisions. And clever advertisers and clever decision makers will frame things in different ways to give you — give rise to different intuitions. Sometimes this could be fairly simple. So, you have this ad of a hamburger that's eighty percent fat free versus twenty percent fat — You don't have to be a brilliant ad executive to figure out which one to go for.

It turns out that this sort of fundamental act – the fundamental role of framing effects – is not limited to humans. So, I want to take a second and tell you some work done by my colleague, Laurie Santos, with capuchin monkeys. And what she does is she takes these capuchin monkeys and she teaches them to use money. She teaches them to use little discs to buy themselves either pieces of banana or pieces of apple. And they like to eat this. And they very quickly learn you can hand over a disc to get some banana or some apple. [laughter] Now, Dr. Santos and her colleagues have done many studies using this method, but the study I'm interested in illustrating here shows framing effects in these nonhuman primates.

So, what she does is — There's two options. In one option, the experimenter shows one object to the capuchin and low — and then either gives one or two — half the time gives one, half the time gives two, for an average of one and a half. The other experimenter does exactly the same thing; gives one or two for an average of one and a half, but starts off displaying two. Now, if you weren't a human, how would you feel about these two experimenters? They both give you the same amount. And capuchins are extremely sensitive to how much they get, but it turns out as predicted they don't like the pink experimenter because the pink experimenter is — he gives you two — shows you two and half the time he gives you one. This guy shows you one, and half the time gives you two. And over time they develop a preference for the experimenter that shows them one initially, suggesting that they are being subject to framing effects or choices relative to a reference point.

A different sort of demonstration is the "endowment effect." This is a robust and very interesting effect. Here's the idea. I show you something like a cup or a chocolate bar and I say, "How much will you give me for this chocolate bar? It looks like you're pretty hungry. How much will you give me for this chocolate bar?" And you say, "I'll give you two dollars for this chocolate bar." Most people on average

give two — the chocolate bar — gives two dollars for a chocolate bar. The other condition's exactly the same except I hand you a chocolate bar and say, "How much money will you sell me that chocolate bar for?" There, people say, "Two fifty," and in fact, what happens is once you own something its value shoots up. And this has mystified economists and psychologists. It makes no sense. The chocolate bar doesn't even have to move. I just leave it on the table and say either "How much will you spend," "How much will you give me for this?" or "Okay. It's yours. How much do you want for me to take it back?" The answer is, it's framing. If you're asking how much you want for it, it's a game. It's just how much will you pay to get something. But if you're being asked how much do you want for me to take it from you, you treat it as a loss. And as a loss it becomes more valuable. Those are framing effects.

The second example is base rates. There are seventy lawyers — sorry, seventy engineers and thirty lawyers and John is chosen at random. Let me tell you about John: forty-years old, married, three children, conservative, cautious, no interest in politics, awkward around people. His hobbies include carpentry, sailing, and solving mathematical puzzles, like online dating. [laughter] What do you think John is? A lawyer or an engineer? Who thinks he's a lawyer? Good. Who thinks he's an engineer? Okay. Most people think he's an engineer, but here's the thing. You switch it. Right? Thirty engineers, seventy lawyers? It doesn't change. People — No matter what this number is — these numbers — it doesn't seem to change who you think he is or how confident you are.

People look at John as an individual and they ignore the background status of where he came from. They ignore base rates. Base rates are very difficult to think about and I want to give you an example of this. And the example will be on the slides for when you print them out — print it out because you might want to work through it yourself. But I'll give this to you quickly.

There's a disease that hits one in a thousand people, a pretty common disease. There's a test for the disease and if you have it, it's going to tell you you have it. It tests for a certain thing in your blood and "boom," if the thing is in your blood the test will go "boom." If you have it, it will tell you you have it. It doesn't miss. On the other hand, it's not perfect. It has a false positive rate of five percent. So, if you don't have the disease, five percent of the time the test will say you have it. So, if the test says you don't have it, you're fine. But if the test says you have it, maybe you have it but maybe it's a false positive. You take the test. It says you have the disease. Without pen and paper, how likely do you think the odds are you have the disease? Who says over fifty percent? Okay. Before people sinisterly shouted the right answer, people will tend — medical students were given this, medical students less savvy than you, and the average is between fifty percent and ninety-five percent.

The answer is, as some people quickly noted, two percent. And here's how it works. One percent of a thousand will have the disease. That person will test positive. The test never misses. That leaves nine hundred ninety-nine people who don't have the disease, and we'll say about fifty percent of these people have it. So, for every fifty-one people who test positive, only one will have the disease, giving an average of about two percent. This sort of thing is very difficult. Our minds are not evolved to do base rate computation. And so, any problems involving base rate computation, including real world problems, like what to do when you come back with a positive test, we screw up. And often we screw up in the direction of panic.

The third bias is the "availability bias." And this is simply that if you want to know how frequent something is, how available it is to come to mind is an excellent cue. But this could lead to mistakes. A

classic example by Kahneman and Tversky is you ask people — one group of people how many English words end with "ng" or what proportion of English words, another group of people what proportion end with "ing." It turns out you get much bigger numbers for "ing" than "ng" though, of course "ng" has to — "ing" — sorry, "ng" would include everything with "ing." It's just a lot easier to think about these things.

This can show up in the real world. What are your risk of getting killed — What's your risk of getting killed by a shark? Well, if you ask people what their risk of getting killed by a shark is, they characteristically overestimate it. I will give you the news of what the risk is for getting killed by a shark. Injured in any given year: one in six million. Killed: one in five hundred million. If you live in Florida, which apparently is Shark Central, your chance of getting injured is about one in a half million. People will overestimate the risks because shark attacks are very salient. They are always reported in the news and they're very interesting. What is the chance of getting killed by potato salad? [laughter] Well, food poisoning, death by food poisoning, injury by food poisoning runs to about one in fifty-five, one in 800 for some sort of injury and one in 55,000 killed. Potato salad is 1,000 more times more dangerous than shark attacks. But you get it wrong because you don't think, "Oh, my God, big news story. Somebody dies by potato salad." [laughter] And so, we tend to overestimate the chance of being killed by dramatic effects.

How many Jews in the United States, what proportion? Who thinks it's over three quarters of the United States is Jewish? [laughter] I'm kind of anchoring here. Okay. Okay. Who thinks over half? Who thinks over forty percent? Who thinks over twenty percent? Okay. Who thinks over fifteen percent? Who thinks over ten percent? Who thinks over seven and one-half percent? Who thinks over five percent? Okay. Who thinks overall there's more than five percent of the United States that's Jewish? Who thinks over three percent? The answer is somewhere between 1.9 and 2.1%. Most people think — The average American thinks it's twenty percent. There is — [laughter] If you're curious about demographics, and this map isn't to be entirely trusted because I got it from Wikipedia, [laughter] this is the distribution of the Jewish population, self-identified as Jewish in different parts of the United States. [laughter] New York City is, of course, the most dense population with nine percent. New Haven has 3.5%.

Now, why do people get it wrong? Well, there's all sorts of reasons and this is going to come out in the context of social psychology when we talk about how people think about human groups. But one quick answer is people who are plainly Jewish are prominent in positions where people notice them, like entertainment or, in the case of you guys, academia. And this could lead to — this availability — "Can I think of a Jew? Yeah." [laughter] This availability causes us to overestimate the proportion to which Jews are represented in the population.

Final example. Confirmation bias. This is a very nice study and it's very simple. It's — You're in a jury of a custody case. You have to give a child custody – either a mother or father sole custody. One parent has average income, average health, average working hours, reasonable rapport with the child, and a relatively stable social life. The second parent has an above-average income, minor health problems, lots of work-related travel, a very close relationship with the travel — with the child, and an extremely active social life. Think for a moment. Who would you award custody with? There's no — Obviously, there's no right answer here.

Just think for a moment. Who would award custody to parent A? Who would award custody to parent B? Okay. As I think there is in this room, when this study is done there's a slight advantage to parent B. Here's what's interesting. You give another group of people this question. "Which parent would you deny custody to?" You get a slight advantage for parent B. Now, this is to some extent an illustration of framing problem but it's also a more general illustration of the confirmation bias. So, when you're asked to award custody to, you then ask, "Well, what is a good — what is a sign that somebody's a good parent?" And the good parent aspects of B jump out. When asking about denying custody you ask, "Where is a cue that somebody's a bad parent?" And the bad parent aspects of B jump out. In general, when we have a hypothesis we look for confirmations.

This makes some things, which are logically easy extremely difficult problems when we face them in the real world. And I'll end with my final example, that of the Wason selection task. Here's the game. And people — I don't want people to shout it out just yet. There is four cards [pointing to a slide depicting four cards: D, G, 3, 8]. Each card has a letter on one side and a number on the other side. You have to judge whether this claim is true or false. "If a card has a 'D' on one side, it has a '3' on the other side." How many cards do you have to turn over to test whether that rule is right? Okay. Somebody shout out what one card is you have to turn over. "D." Everybody gets that right. What else? Do you need to do any other cards? How many people think it's "D" and "3"? I'm raising my hand to fool you. [laughter] People answer either "D" or "D" and "3" but think about it. What would make this rule wrong? It's wrong if it has "D" on one side and not "3" on the other. Right? That's what it would be to be wrong. You then would have to check "D" to see if there is a "3" on the other side. You were all right about that. That means you'd check "8" to see if there's a "D" on the other side. "Three's" not going to tell you anything. That's hard. People find this very hard.

Okay. Big deal. But what's interesting is you can modify it in certain ways to make it a lot easier. And this is the work of Leda Cosmides and her colleague, an evolutionary psychologist at Santa Barbara who has argued that if you frame these questions in ways that make ecological sense, people are much better at them. And basically, she does studies where she has people who are evaluating a social rule. Imagine these cards. On one side of the card is an alcohol — is a drink. On the other side is a person's age. You are a bartender and you want to make sure nobody under twenty-one drinks beer. Which cards do you turn over? Well, now it's easier but the logic is the same. It's a violation that there's "under twenty-one" on one side, "beer" on the other side, so you need to check the "under twenty-one" here and you need to check the "beer" here. And when you make these logical problems more ecologically valid they turn out to be much easier.

Okay. There's a little bit more but I'll hold it off until next class. And I'll end with the reading response, which is to do your own bit of reverse engineering and evolutionary psychology. And I'll see you all on Wednesday.